Sleep Apnoea: An Approach to a Case of Fatigue

Dr. Essam Kotb, Internal Medicine | Source: `13- sleep apnea updated.pdf` (30 slides)

Legend: ⭐ = high-yield · *[added]* = not on slides. Images marked "Slide N" come from the lecture; "Added diagram" = drawn for these notes. The case answer is **reasoned by me; the key is not on the slides**. "Case Scenario 2" (slide 28) is only a cartoon, with no question.

🔗 **Bridge from earlier lectures:**

- **Pulmonary vascular disease:** OSA/OHS = **WHO PH group 3** (hypoxia) → **cor pulmonale**.
- **Hypertension:** OSA is a classic **secondary cause of resistant hypertension**.
- **Arrhythmias:** OSA → **AF** and nocturnal brady-arrhythmias.
- **Pituitary I / Thyroid: acromegaly** (macroglossia) and **hypothyroidism** are listed OSA risk factors.
- **Restrictive lung disease:** obesity is an extraparenchymal restrictive cause; OHS is its ventilatory failure.
- **Up next (Respiratory failure):** OHS = **type 2 (hypercapnic) failure**, often first diagnosed in acute decompensation.

1. Title / Objectives

Domain	Objective	Likely tested as
Knowledge	Define **OSA, CSA, OHS (Pickwickian)**; normal sleep stages	Definitions
Knowledge	Mechanisms: airway collapse (OSA), respiratory-centre failure (CSA), hypoventilation (OHS)	
Knowledge	⭐ Nocturnal hypoxia/hypercapnia → **pulmonary vasoconstriction → PH → cor pulmonale**; secondary HTN; renal compensation	Lecture case 1
Knowledge	Risk factors and features: snoring, sleepiness, **morning headache**, witnessed apnoeas, **Cheyne–Stokes**	
Skill	⭐ **STOP-Bang** screening	Score ≥ 5 = high risk
Skill	⭐ **Polysomnography**: obstructive vs central events	Effort present vs absent
Skill	Choose therapy: lifestyle → **CPAP** → NIPPV/surgery	CPAP first line
Values	Professionalism; impact of fatigue and sleepiness on quality of life and **safety** (driving); multidisciplinary care	

Main subtopics:

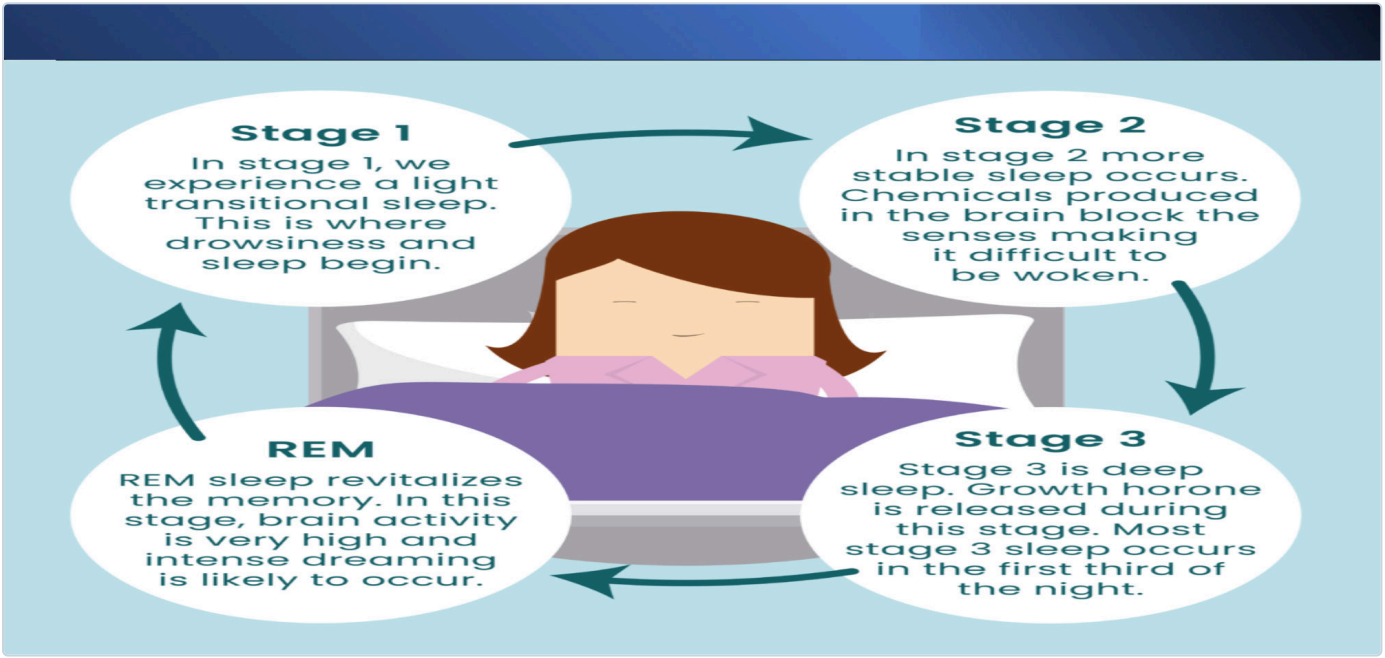
1. **Obstructive sleep apnoea**
2. **Central sleep apnoea**
3. **Obesity hypoventilation syndrome**

2. Main Content (Lecture Order)

2.1 Sleep and Its Stages (Slides 5–6)

Sleep is a natural state of rest in which the body and mind recover: tissue repair and regeneration, memory consolidation, hormonal balance, energy, mood, immunity.

Stage	Features
Stage 1	Light, **transitional** sleep; drowsiness, sleep onset
Stage 2	More stable sleep; brain chemicals **block the senses** so waking is harder
⭐ **Stage 3**	**Deep (slow-wave) sleep**; ⭐ **growth hormone released**; mostly in the **first third** of the night
⭐ **REM**	Memory revitalisation; **high brain activity**; **intense dreaming** (*[added]* skeletal-muscle atonia, so OSA and OHS hypoventilation are **worst in REM**)



Slide 6: Sleep stages

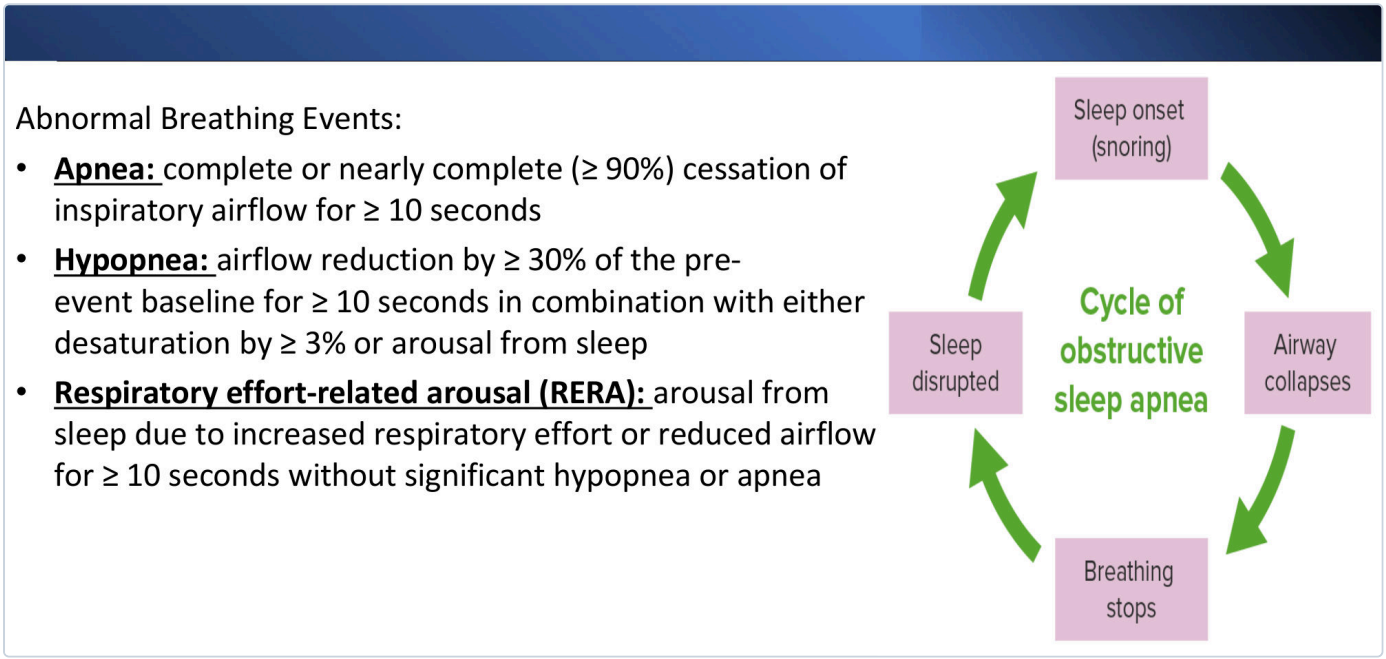
2.2 Sleep Disorders and Abnormal Breathing Events (Slides 7–9)

- **Types of sleep disorders:**
  - insomnia
  - ⭐ **sleep-related breathing disorders**
  - central disorders of hypersomnolence
  - circadian rhythm sleep–wake disorders
  - parasomnias
  - sleep-related movement disorders
- **Sleep-disordered breathing (slide 8):** snoring · **OSA** · **upper airway resistance syndrome** · **CSA**.

Event	Definition
⭐ **Apnoea**	**Complete or near-complete (≥ 90%) cessation** of inspiratory airflow for **≥ 10 s**
⭐ **Hypopnoea**	Airflow **↓ ≥ 30%** from baseline for **≥ 10 s plus either ≥ 3% desaturation or an arousal**

Event	Definition
RERA (respiratory effort-related arousal)	**Arousal** from ↑ respiratory effort or ↓ airflow for **≥ 10 s without** meeting apnoea/hypopnoea criteria

- **Cycle of OSA (slide 9):** sleep onset (snoring) → airway collapses → breathing stops → sleep disrupted → back to sleep.
- *[added]* **AHI** (apnoea–hypopnoea index) = apnoeas + hypopnoeas per hour of sleep.
  - **Mild 5–14 · moderate 15–29 · severe ≥ 30.**
  - **OSA diagnosis:** AHI **≥ 15**, or **≥ 5 with symptoms or comorbidity** (sleepiness, HTN, AF, stroke, mood disorder).
  - The hypopnoea rule above is the AASM "3% or arousal" rule; some payers use **≥ 4% desaturation**.



Slide 9: Breathing events; cycle of OSA

2.3 Obstructive Sleep Apnoea: Definition and Aetiology (Slides 10–11) ⭐

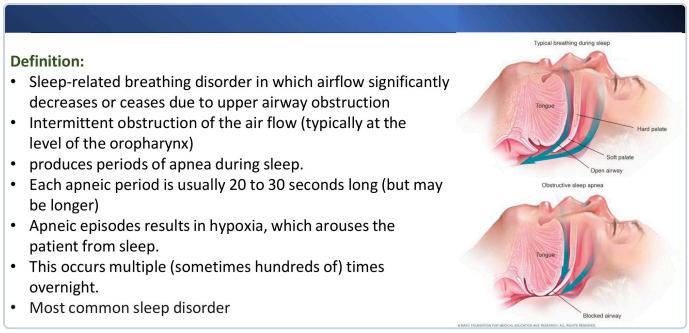
Definition:

- airflow **decreases or ceases** due to **upper airway obstruction**
- **intermittent**, typically at the **oropharynx**
- each apnoea usually **20–30 s** (may be longer)
- **hypoxia arouses the patient**; repeated **many (sometimes hundreds of) times** a night
- ⭐ **the most common sleep disorder**

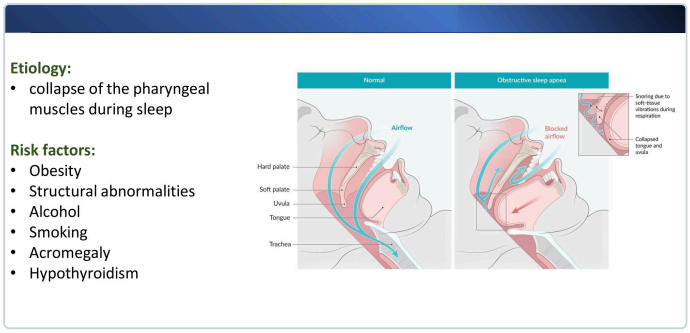
Aetiology: ⭐ collapse of the pharyngeal muscles during sleep.

Risk factors:

- ⭐ **Obesity**
- **Structural abnormalities** (*[added]* tonsillar hypertrophy, retrognathia, macroglossia)
- **Alcohol** (and sedatives)
- **Smoking**
- ⭐ **Acromegaly**
- ⭐ **Hypothyroidism**



Slide 10: Open vs obstructed airway



Slide 11: OSA airway anatomy

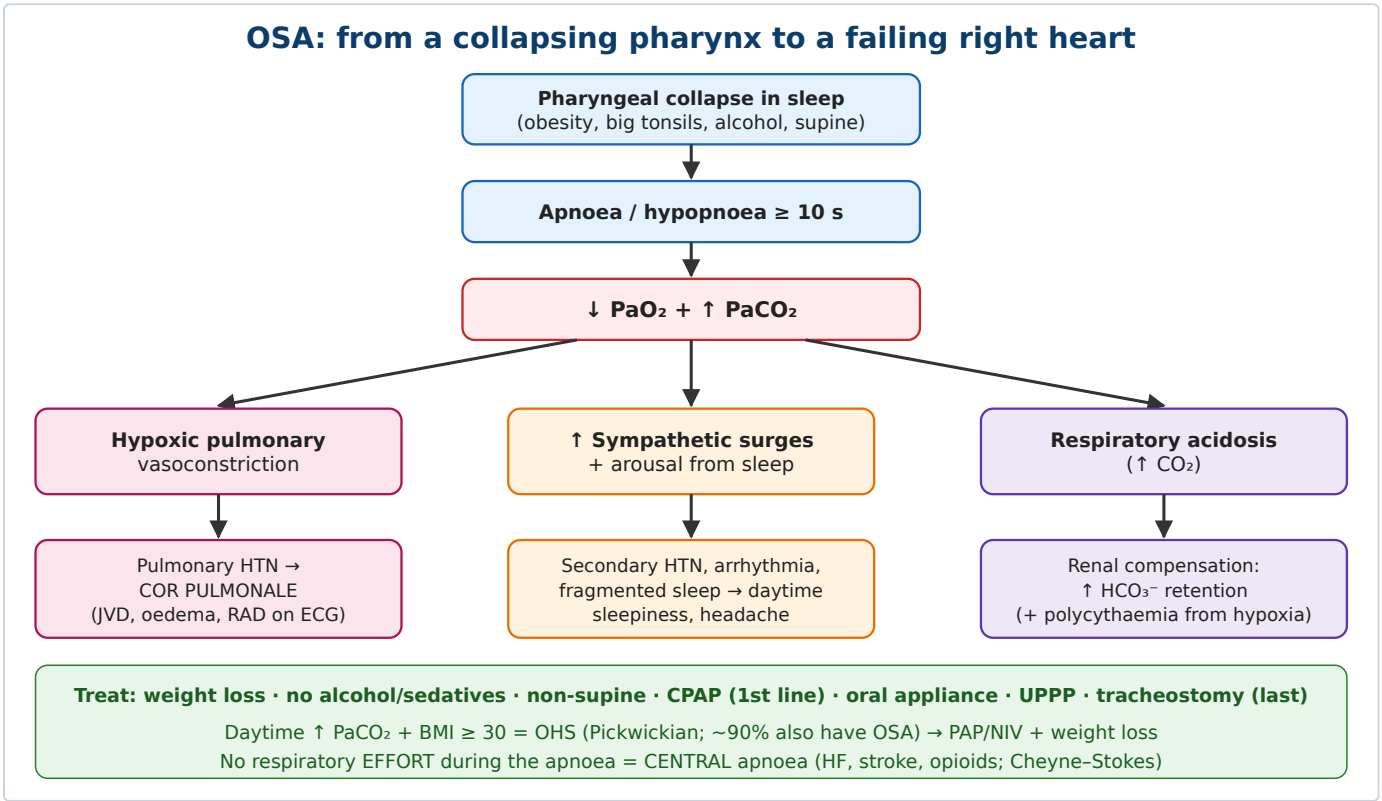
2.4 OSA: Pathophysiology (Slide 12) ⭐⭐

Upper airway obstruction → apnoea → **↓ PaO_2 and ↑ $PaCO_2$**, which lead to:

1. ⭐ **↑ Hypoxic pulmonary vasoconstriction → ↑ pulmonary hypertension → cor pulmonale**
2. ⭐ **↑ Sympathetic activity → secondary hypertension**
3. ⭐ **Respiratory acidosis → renal compensation → ↑ HCO_3^- retention**

Venn of contributing factors:

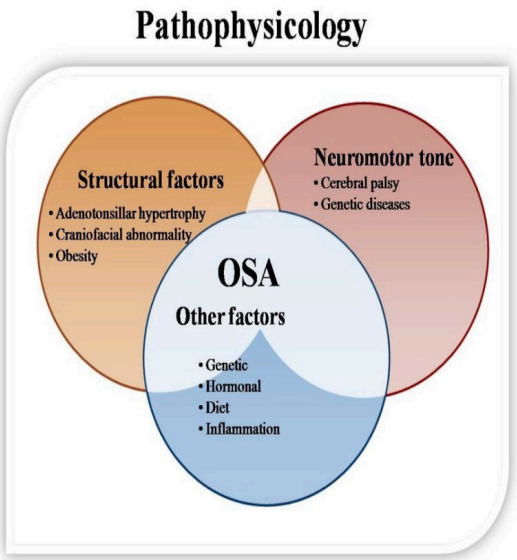
- **structural** (adenotonsillar hypertrophy, craniofacial abnormality, obesity)
- **neuromotor tone** (cerebral palsy, genetic diseases)
- **other** (genetic, hormonal, diet, inflammation)



Added diagram: OSA cascade from pharyngeal collapse to cor pulmonale

Pathophysiology

- Obstruction of the upper airway → apnea → ↓ PaO2 &↑ partial PaCO2 which leads to:
- ↑ Hypoxic pulmonary vasoconstriction → ↑ pulmonary hypertension → cor pulmonale
- ↑ Sympathetic activity → secondary hypertension
- Respiratory acidosis → renal compensation → increased HCO3 retention



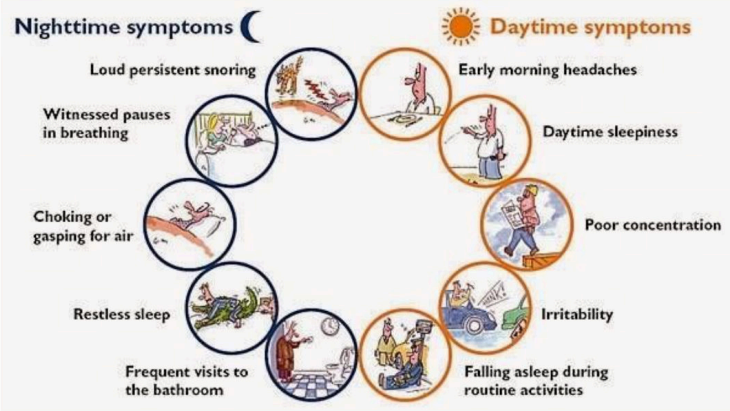
Slide 12: OSA pathophysiology

2.5 OSA: Clinical Features (Slide 13) ⭐

Night-time	Daytime / systemic
⭐ **Loud, persistent snoring**	⭐ **Daytime sleepiness**; falling asleep during routine activities (*[added]* driving accidents)
⭐ **Witnessed pauses in breathing**	⭐ **Early-morning headaches** (CO_2)
Choking/gasping for air	**Personality changes**, **↓ intellectual function**, poor concentration, irritability
Restless sleep	**Pulmonary HTN**, **cardiac arrhythmias**, **hypertension**
Nocturia (frequent bathroom visits)	⭐ **Polycythaemia**
—	**↓ Libido**

Clinical Features:

- Snoring
- Daytime sleepiness
- Personality changes,
- Decreased intellectual function
- Pulmonary HTN
- Cardiac arrhythmias
- Morning headaches
- Polycythemia
- Hypertension
- Decrease Libido



Slide 13: OSA night and day symptoms

2.6 OSA: Diagnosis (Slides 14–15) ⭐

Steps:

1. **Sleep history, including third-party reports** (bed partner)
2. evaluate comorbidities
3. **screen with a standardised questionnaire**
4. ⭐ **polysomnography is required to confirm**

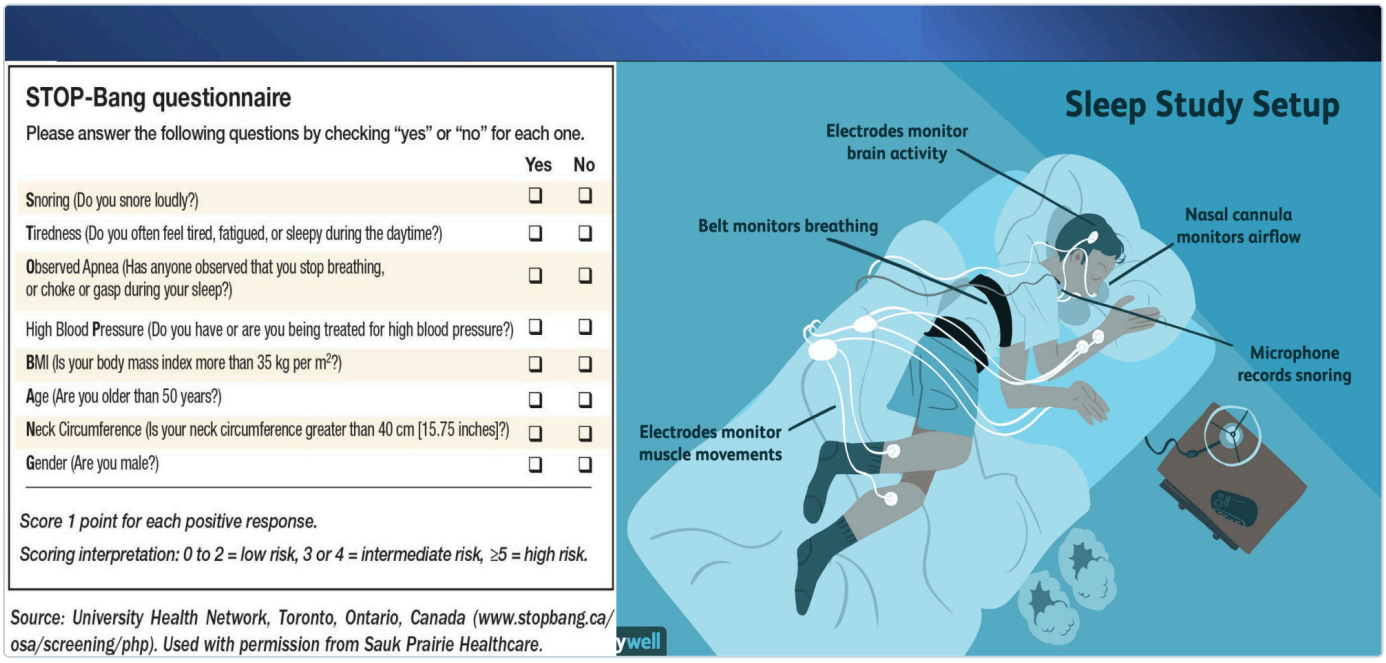
⭐ **STOP-Bang questionnaire** (1 point each):

Letter	Question
S	**Snoring** loudly?
T	**Tired**/sleepy in the daytime?
O	**Observed** apnoea, choking or gasping?
P	High blood **Pressure** (or treated)?
B	**BMI > 35** kg/m²?
A	**Age > 50**?
N	**Neck circumference > 40 cm**?
G	**Gender** male?

⭐ **Score: 0–2 = low risk · 3–4 = intermediate · ≥ 5 = high risk.**

Polysomnography (sleep study) setup:

- electrodes for **brain activity** (EEG), **eye** and **muscle movements**
- **belts** for respiratory effort
- **nasal cannula** for airflow
- **microphone** for snoring
- *[added]* **pulse oximetry**, ECG
- *[added]* **Home sleep apnoea testing** is acceptable for uncomplicated suspected moderate–severe OSA.
- *[added]* **Epworth Sleepiness Scale** (> 10) quantifies daytime sleepiness.



Slide 15: STOP-Bang and sleep study setup

2.7 OSA: Management (Slide 16) ⭐⭐

Level	Intervention
Lifestyle	⭐ **Weight loss and exercise** · **stop smoking** · **avoid alcohol and sedatives** · **non-supine sleeping**
Device	**Oral appliance (mandibular advancement)**: keeps the jaw forward, prevents airway collapse
⭐ **First line**	⭐ **CPAP** (pneumatic splint)
Surgery	**Uvulopalatopharyngoplasty (UPPP)** in **selected** patients
Last resort	**Tracheostomy** (severe disease)

[added]

- **Hypoglossal nerve stimulation** for CPAP-intolerant moderate–severe OSA.
- **GLP-1 agonist tirzepatide** is now approved for OSA with obesity.
- Advise about **driving** (notify the licensing authority if sleepy).

Management:

- weight loss and exercise
- Stop smoking
- avoiding alcohol and sedatives
- Oral appliance maintain mandibular advancement and prevent airway collapse
- sleeping in a nonsupine position
- **Continuous positive airway pressure (CPAP) first line**
- Uvulopalatopharyngoplasty (selective patients)
- Tracheostomy (last resort in cases of severe disease)



Slide 16: OSA treatments; mandibular advancement

🔁 **Recap: OSA.**

- Pharyngeal collapse at the oropharynx; apnoea ≥ 10 s (≥ 90% airflow loss); hypopnoea ≥ 30% + 3% desaturation or arousal.
- Risks: obesity, structural, alcohol, smoking, **acromegaly, hypothyroidism**.
- ↓O_2/↑CO_2 → pulmonary vasoconstriction (cor pulmonale), sympathetic surges (HTN, arrhythmia), acidosis (↑ HCO_3^-), polycythaemia.
- Snoring, witnessed apnoeas, sleepiness, morning headache.
- **STOP-Bang ≥ 5 = high risk; PSG confirms.**
- **CPAP first line**; weight loss, avoid alcohol, positional therapy, oral appliance; UPPP selected; tracheostomy last.

▶️ **Up next: Central sleep apnoea.**

2.8 Central Sleep Apnoea (Slides 17–21) ⭐

Definition: the **drive to breathe periodically decreases or ceases** due to **impaired respiratory-centre function**. In OSA, by contrast, the drive is intact but the airway is blocked.

Mechanism	Hypocapnic CSA (hyperventilation)	Hypercapnic CSA (hypoventilation)
Category / prevalence	**Primary CSA** (prevalence low) · ⭐ **Cheyne–Stokes breathing** (prevalence **high**)	**Obesity · neuromuscular disease · ⭐ opiates · neurodegenerative** (stroke, multisystem atrophy, Parkinson's, Alzheimer's); prevalence unknown
Treatment	**CPAP, acetazolamide, O_2, CO_2, theophylline**	⭐ **NIPPV with a backup rate**

Osmosis "3 C's" (slide 21):

- **C**NS injury or toxicity
- ⭐ **C**ongestive heart failure
- ⭐ **C**heyne–Stokes respiration

Signs and symptoms (slide 19): similar to OSA, plus ⭐ **Cheyne–Stokes breathing**: a **crescendo–decrescendo** pattern alternating with apnoea.

Polysomnography (slide 19):

- ⭐ **no airflow AND no respiratory effort** (thoracic and abdominal belts flat)
- **desaturation** follows each apnoeic episode

⭐ **The key differentiator:**

Apnoea type	Respiratory effort
Obstructive	**Present** (chest moves against a closed airway)
Central	**Absent**

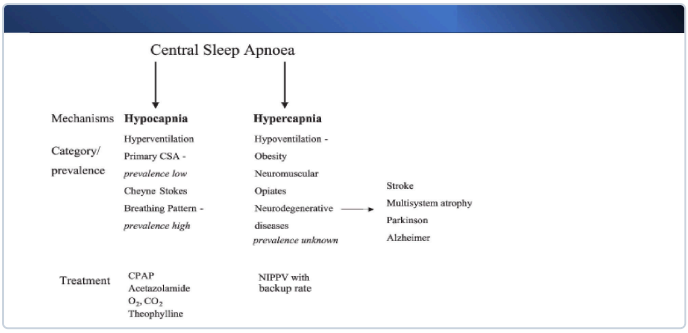
Diagnosis (slide 20):

- **history of underlying conditions (HF, stroke)**
- **polysomnography**, ideally with ⭐ **oesophageal pressure manometry (the gold standard** for measuring effort)

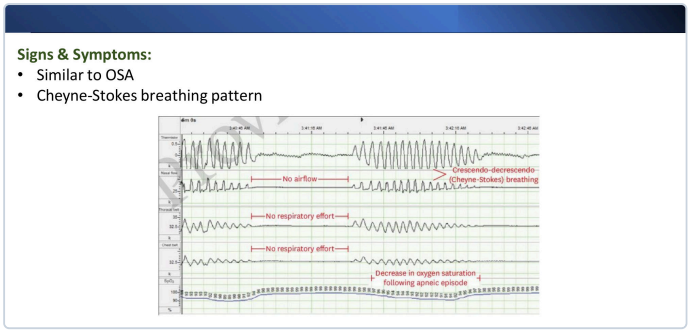
Management (slide 21):

- ⭐ **positive airway pressure therapy**
- **treat the underlying cause** (optimise HF; reduce opioids)

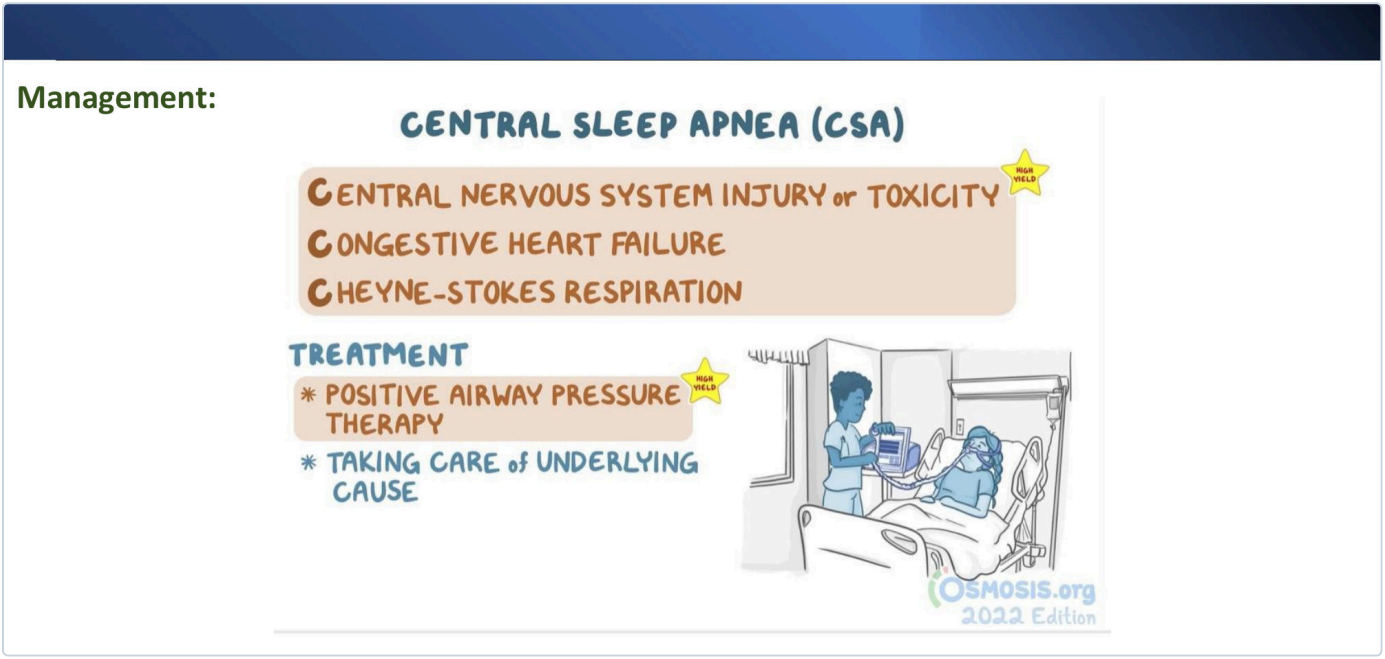
⚠️ *[added]* **Adaptive servo-ventilation (ASV) is contraindicated in HFrEF (EF ≤ 45%)** with central apnoea (SERVE-HF: ↑ mortality).



Slide 18: CSA, hypocapnic vs hypercapnic



Slide 19: Cheyne–Stokes on polysomnography



Slide 21: CSA, causes and treatment

2.9 Obesity Hypoventilation Syndrome (Pickwickian) (Slides 22–25) ⭐⭐

Definition: a sleep-related hypoventilation disorder defined by:

1. ⭐ **BMI ≥ 30 kg/m²**
2. ⭐ **DIURNAL hypercapnia** (*[added]* awake **$PaCO_2$ > 45 mmHg**)
3. **disordered breathing during sleep**

[added] after **excluding other causes** of hypoventilation (COPD, neuromuscular disease, hypothyroidism, chest-wall disease).

Component	Mechanism
Structural	**↓ Inspiratory muscle strength** + **restricted respiratory excursion** → alveolar hypoventilation (fat impairs lung expansion and restricts movement)
Central	**↑ Work of breathing → ↑ drive → can't be maintained in REM** → sleep hypoventilation → **blunting of the central respiratory centres → diurnal hypercapnia**
Obstructive	⭐ **~90% of OHS patients also have OSA**

Clinical features:

- **similar to OSA**: **headaches, severe sleepiness**
- **possibly daytime dyspnoea**

- ⭐ **often diagnosed in an acute state of respiratory failure** (*[added]* acute-on-chronic hypercapnic failure; ↑ HCO_3^- is a clue)

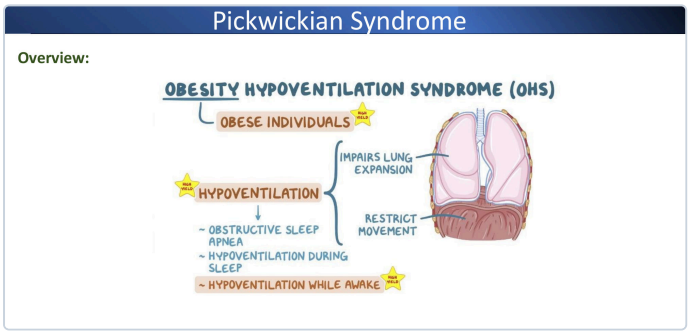
Blood gases (slide 25):

- ⭐ **↑ $PaCO_2$ while AWAKE and SLEEPING**
- **↓ PaO_2 while sleeping**

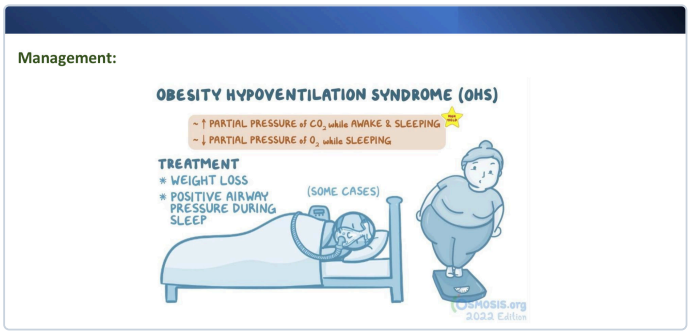
Management:

- ⭐ **weight loss**
- ⭐ **positive airway pressure during sleep**
- *[added]* **CPAP** if severe OSA coexists; **NIV/BiPAP** if no severe OSA or CPAP fails
- *[added]* avoid sedatives; **cautious O_2** (CO_2 retention); bariatric surgery in selected patients

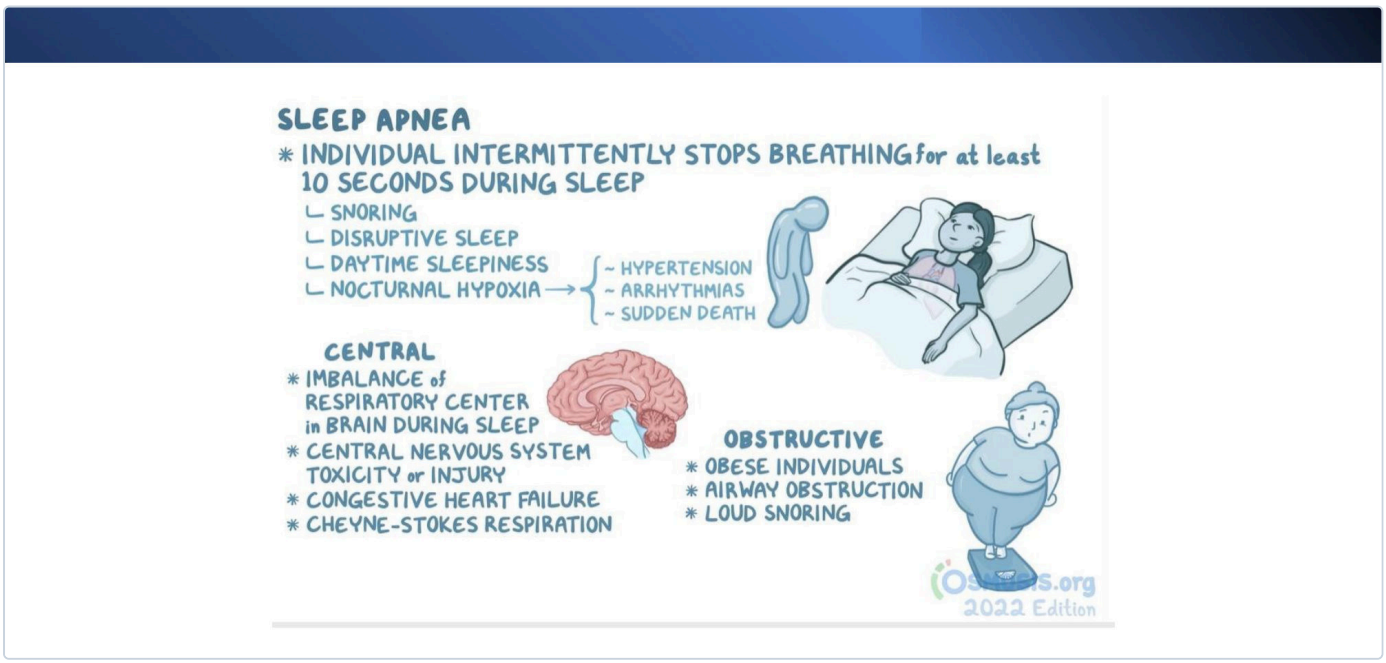
	OSA	**OHS**
Daytime $PaCO_2$	Normal	⭐ **↑ (> 45)**
BMI	Often obese	**≥ 30 required**
Serum HCO_3^-	Usually normal	↑ (*[added]* > 27 mmol/L is a screening clue)
Treatment	CPAP	PAP (CPAP or **NIV**) + weight loss



Slide 22: OHS overview



Slide 25: OHS management



Slide 26: Summary

🔁 **Recap: CSA and OHS.**

- **CSA:** no drive, no effort.
- Hypocapnic (Cheyne–Stokes in **HF**, primary) → CPAP/acetazolamide/O_2.
- Hypercapnic (opiates, neuromuscular, obesity, neurodegenerative) → **NIPPV with a backup rate**.
- Diagnose with PSG ± oesophageal manometry.
- **OHS:** BMI ≥ 30 + **daytime hypercapnia** + sleep-disordered breathing; 90% have OSA; often presents in acute respiratory failure; weight loss + PAP.

---

3. High-Yield Comparison Tables

Table A: OSA vs CSA vs OHS

	OSA	CSA	OHS
Problem	Airway **collapse**	**No drive**	**Hypoventilation** (obesity)
Effort during apnoea	⭐ **Present**	⭐ **Absent**	Reduced
Pattern	Snoring, gasping	**Cheyne–Stokes** (HF, stroke)	Shallow breathing; worst in REM
Daytime $PaCO_2$	Normal	Low (Cheyne–Stokes) or high (opiates, neuromuscular)	⭐ **High**
Key risks	Obesity, tonsils, alcohol, acromegaly, hypothyroidism	**HF**, stroke, **opioids**, neurodegeneration	**BMI ≥ 30**
First-line treatment	⭐ **CPAP**	**PAP + treat cause**; NIPPV with backup rate if hypercapnic	**Weight loss + PAP/NIV**

Table B: Breathing Events

Event	Airflow	Duration	Additional criterion
Apnoea	**≥ 90% ↓**	≥ 10 s	—
Hypopnoea	**≥ 30% ↓**	≥ 10 s	**≥ 3% desaturation or arousal**
RERA	Limited	≥ 10 s	**Arousal**, not meeting the above

Table C: OSA Consequences

Consequence	Mechanism
Cor pulmonale	Hypoxic pulmonary vasoconstriction → PH
Secondary HTN	Sympathetic activation
↑ HCO_3^-	Renal compensation of respiratory acidosis
Polycythaemia	Chronic hypoxia → EPO
Arrhythmia	Hypoxia + sympathetic surges
Morning headache	Nocturnal CO_2 retention
Sleepiness, cognition, mood	Sleep fragmentation

---

4. Rules, Exceptions, Patterns

- **Apnoea ≥ 10 s, ≥ 90% airflow loss**; **hypopnoea ≥ 30% + 3% desaturation or arousal**.
- **OSA is the most common sleep disorder**; obesity is the main risk factor.
- **Acromegaly and hypothyroidism** are endocrine causes of OSA (macroglossia, myxoedema).
- **STOP-Bang: B = BMI > 35, A = age > 50, N = neck > 40 cm**; score ≥ 5 = high risk.
- **PSG confirms**; history is not enough.
- **CPAP is first line**; tracheostomy is the last resort.
- **Effort present = obstructive; effort absent = central.**
- **Cheyne–Stokes = HF/stroke** (hypocapnic CSA).
- **Opiates → hypercapnic CSA** → NIPPV with a backup rate.
- **OHS requires DAYTIME hypercapnia**; 90% also have OSA; often first seen in acute respiratory failure.
- *[added]* **ASV is contraindicated in HFrEF.**

---

5. Clinical Correlations

- **Resistant hypertension** → screen for OSA (with aldosteronism, renal artery stenosis; HTN lecture).
- **AF recurring after ablation/cardioversion** → think OSA.
- **Unexplained polycythaemia** or **↑ HCO_3^-** in an obese patient → OSA/OHS.
- **Right heart failure with clear lungs and RAD** in a snorer → OSA → PH → **cor pulmonale** (lecture case 1).
- **Acromegaly / hypothyroidism** → OSA: check before anaesthesia.
- **Perioperative:** OSA patients are at risk with opioids and sedatives; bring their own CPAP *[added]*.
- **Driving / occupational safety:** daytime sleepiness → accidents.
- **HF clinic:** Cheyne–Stokes CSA marks severe HF; optimise HF therapy.
- **Opioid users:** central apnoea; review doses.

---

6. Rapid Revision (Last-Day)

1. **Sleep stages:**
   - N1 transitional
   - N2 stable
   - **N3 deep (GH; first third of the night)**
   - **REM** (dreams, high brain activity; *[added]* atonia, worst hypoventilation)
2. **Events:**
   - **apnoea ≥ 90% for ≥ 10 s**
   - **hypopnoea ≥ 30% for ≥ 10 s + ≥ 3% desaturation or arousal**
   - **RERA** = arousal without either
3. **OSA:** upper airway (oropharyngeal) obstruction; apnoeas of 20–30 s; hypoxia → arousal; hundreds a night; the **most common sleep disorder**.
4. **OSA risks: obesity**, structural, alcohol, smoking, **acromegaly, hypothyroidism**.
5. **Pathophysiology:** ↓O_2/↑CO_2 → (a) **pulmonary vasoconstriction → PH → cor pulmonale**; (b) **sympathetic → secondary HTN**; (c) **acidosis → ↑ HCO_3^-**.
6. **Features:**
   - snoring, sleepiness, **morning headache**

- personality change, ↓ intellect
- PH, arrhythmia, HTN, **polycythaemia**, ↓ libido
- witnessed apnoeas, nocturia

7. **Diagnosis:** history + partner, comorbidities, **STOP-Bang** (≥ 5 high), **polysomnography** (confirmatory).
8. **STOP-Bang:** Snoring, Tired, Observed, Pressure, **BMI > 35, Age > 50, Neck > 40 cm, Gender male**.
9. **OSA Rx:**
   - weight loss, exercise, stop smoking, no alcohol/sedatives, non-supine
   - oral appliance
   - **CPAP first line**
   - UPPP (selected); tracheostomy (last)
10. **CSA:** impaired respiratory-centre drive. - **Hypocapnic** (primary, **Cheyne–Stokes**) → CPAP, acetazolamide, O_2, CO_2, theophylline. - **Hypercapnic** (obesity, neuromuscular, **opiates**, neurodegeneration) → **NIPPV with a backup rate**.
11. **CSA PSG: no airflow + no effort**. Gold standard: PSG with **oesophageal manometry**. Treat with PAP + the underlying cause (HF, stroke).
12. **OHS:** - **BMI ≥ 30 + daytime hypercapnia** + sleep-disordered breathing - structural, central and obstructive components (**90% have OSA**) - headache, sleepiness, daytime dyspnoea; **often diagnosed in acute respiratory failure** - **↑ $PaCO_2$ awake and asleep; ↓ PaO_2 asleep** - **weight loss + PAP during sleep**

---

7. Exam-Style Questions

Q1 (lecture case 1, slide 27). A 56-year-old man has 5 months of progressive **bilateral ankle swelling** and exertional dyspnoea, with constant daytime tiredness. His wife reports **snoring and choking in sleep**. 25 pack-years; HTN on enalapril. BP 145/95, **JVD**, **lungs clear**, warm peripheries, 2+ oedema, ECG **right axis deviation**. Most likely cause? a. LVH · b. Ischaemic heart disease · **c. Chronic hypoxia** · d. Chronic kidney damage · e. Pulmonary neoplasia · f. Alveolar destruction

> **Answer: c** *[reasoning added; key not on slides]*.
> - Snoring, choking and daytime sleepiness = **OSA**.
> - Recurrent nocturnal hypoxia → **hypoxic pulmonary vasoconstriction → pulmonary hypertension → cor pulmonale**: JVD, oedema, **RAD**, with **clear lungs** (so not left HF).
> - Why not the others:
> - **f (emphysema)** would give hyperinflation and wheeze/↓ breath sounds; the lungs are clear and the sleep history points to OSA.
> - **a/b** cause left-sided failure with crackles.

Q2. A 48-year-old man (BMI 37, neck 44 cm, treated HTN) snores loudly, falls asleep at traffic lights, and his wife has seen him stop breathing. STOP-Bang score and next step? A. 4; reassure · B. 5; start CPAP empirically · **C. ≥ 6 (high risk); polysomnography** · D. 3; weight loss only

> **Answer: C** (slides 14–15).
> - STOP-Bang points: **S, T, O, P, B (> 35), N (> 40), G (male)** = 7 (age < 50 scores 0). **≥ 5 = high risk.**
> - **Polysomnography is required to confirm** before CPAP (first line once confirmed).

Q3. A 70-year-old with **HFrEF (EF 25%)** has periodic crescendo–decrescendo breathing with apnoeas during sleep. Polysomnography shows **absent thoracoabdominal effort** during apnoeas. Diagnosis? A. OSA · **B. Central sleep apnoea with Cheyne–Stokes breathing** · C. OHS · D. Upper airway resistance syndrome

> **Answer: B** (slides 18–21).
> - **No effort = central.**
> - Cheyne–Stokes is the **hypocapnic** CSA of HF.
> - Treat HF + PAP. *[added]* **Avoid ASV in HFrEF.**

Q4. A 45-year-old woman with **BMI 48** is admitted drowsy. ABG on room air: pH 7.32, **$PaCO_2$ 68**, PaO_2 52, **HCO_3^- 34**. Her CXR is clear and she has no lung disease. Diagnosis? A. Acute asthma · B. Pure OSA · **C. Obesity hypoventilation syndrome (acute-on-chronic hypercapnic failure)** · D. Pulmonary embolism

> **Answer: C** (slide 24).
> - BMI ≥ 30 + hypercapnia; **↑ HCO_3^- shows chronicity** (renal compensation).
> - "**Often diagnosed in an acute state of respiratory failure**".
> - Treat with **NIV/PAP**, weight loss, cautious O_2.

Q5. Which is **not** a recognised risk factor for OSA in this lecture? A. Acromegaly · B. Hypothyroidism · C. Alcohol · **D. Hyperthyroidism** · E. Obesity

> **Answer: D** (slide 11).
> - **Acromegaly** (macroglossia, soft-tissue growth) and **hypothyroidism** (myxoedematous tissue, ↓ drive) are the endocrine causes; hyperthyroidism is not.

---

8. Exam Pitfalls & Traps

Trap	Correct thinking
Diagnosing OSA from STOP-Bang alone	❌ A screen only; **polysomnography confirms**
Hypopnoea = any airflow drop	❌ **≥ 30% for ≥ 10 s + ≥ 3% desaturation or arousal**
Apnoea = 5 s pause	❌ **≥ 10 s, ≥ 90% loss**
Central apnoea = obstructive apnoea in the brain	❌ Central = **no respiratory effort**; obstructive = effort against a closed airway
OHS = obese + OSA	❌ Requires **daytime hypercapnia** ($PaCO_2$ > 45)
High-flow O_2 alone for OHS decompensation	❌ Risk of worsening CO_2; **NIV**
CPAP is second line after surgery	❌ **CPAP first line**; UPPP selected; tracheostomy last
Right heart failure + clear lungs → LV failure	❌ Think **PH / cor pulmonale** (OSA, COPD, PE)
Cheyne–Stokes = OSA	❌ **Central** (hypocapnic) apnoea of HF/stroke
ASV for central apnoea in HFrEF *[added]*	❌ **Contraindicated** (SERVE-HF)
Opioid-related central apnoea → CPAP	⚠️ Hypercapnic CSA → **NIPPV with a backup rate**
Sedatives for poor sleep in OSA	❌ Worsen collapse; **avoid alcohol and sedatives**